# On Efficient Language and Vision Assistants for Visually-Situated Natural Language Understanding: What Matters in Reading and Reasoning

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## Abstract

Recent advancements in language and vision assistants have showcased impressive capabilities but suffer from a lack of transparency, limiting broader research and reproducibility. While open-source models handle general image tasks effectively, they face challenges with the high computational demands of complex visually-situated text understanding. Such tasks often require increased token inputs and large vision modules to harness high-resolution information. Striking a balance between model size and data importance remains an open question. This study aims to redefine the design of vision-language models by identifying key components and creating efficient models with constrained inference costs. By strategically formulating datasets, optimizing vision modules, and enhancing supervision techniques, we achieve significant improvements in inference throughput while maintaining high performance. Extensive experiments across models ranging from 160M to 13B parameters offer insights into model optimization.[1]


## 1 Introduction

Recent advancements in integrating Large Language Models (LLMs) with computer vision have led to the creation of sophisticated Language-Vision Assistants. These systems are capable of interpreting text within images, enabling them to excel in complex tasks requiring both visual and textual understanding. Notably, models like GPT-4(V) (OpenAI, 2023) are able to leverage a single foundation model to handle various text-centric tasks. However, these models also face significant challenges related to transparency and accessibility, limiting broader utilization.

Open-source alternatives such as LLaVA (Liu et al., 2023c, 2024b) have emerged to address these issues. However, as these models grow in complexity, concerns about their reproducibility and resource efficiency persist (Kim et al., 2023; Dong et al., 2024). Some open-source models provide only the model weights without comprehensive specifications, making replication and use more challenging.

In the fast-evolving realm of Vision-Language Models (VLMs), simply expanding model size and consuming more resources does not necessarily enhance practical utility. It is crucial to strike a balance between high performance and resource efficiency to democratize access to advanced VLMs. Particularly, inference costs are a significant concern for practitioners developing real-world appli-

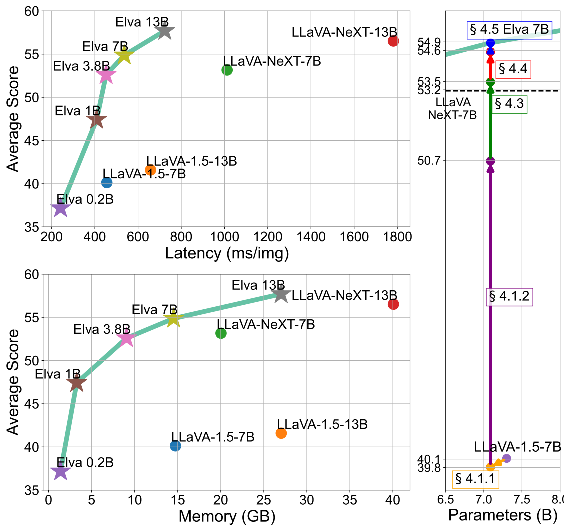


Figure 1: Graphical comparison illustrating average score against latency and memory consumption for various models. Scores are derived from eight benchmarks: DocVQA (Mathew et al., 2021), ChartQA (Masry et al., 2022), InfographicVQA (Mathew et al., 2022), SEED (Li et al., 2023), SEED-2-Plus (Li et al., 2024a), MMStar (Chen et al., 2024), ScienceQA (Lu et al., 2022), and HallusionBench (Guan et al., 2024). See Section 4.2 for benchmark details. **Elva** excels with high performance, reduced latency, and lower memory usage. Right: Performance improvements from LLaVA to Elva, achieved through strategies in Section 4.

[1]We will fully open-source our work.

cations. Despite the importance of this balance, the fundamental elements that contribute to VLM success remain underexplored.

Traditionally, to enhance text comprehension, many VLMs increase their model resolution, often leading to larger and more resource-intensive models. In this work, we challenge this approach by introducing ELVA (**E**fficient **L**anguage and **V**ision **A**ssistant), a suite of VLMs designed to maintain high performance while reducing inference costs and focusing on text-centric tasks. While we do increase training costs to a manageable extent, the primary research target of ELVA is to create models capable of handling high-resolution tasks with low inference costs.

Our key contributions are as follows:

1. **Efficiency and Reproducibility:** We present ELVA, an efficient and scalable model architecture trained on open-source data, demonstrating superior reproducibility and cost-effectiveness as shown in Figure 1.

2. **Empirical Validation:** We conduct thorough experiments to validate the effectiveness of ELVA's primary components.

3. **Model Scalability:** We develop ELVA versions ranging from 0.2B to 13B parameters, showcasing its scalability and adaptability.

4. **Dataset Contributions:** To evaluate ELVA as a document assistant, we introduce two new datasets, CORD-Instruct and Parsing-Bench.

5. **Open-Source Initiative:** To foster further community research and ensure model reproducibility, we will open-source the trained models and datasets from this study.

Our ultimate goal is to shed light on the complexities of VLMs, helping readers identify the critical factors driving model success while presenting a practical, cost-effective solution with significant real-world applications. Following this introduction, §2 provides an overview of foundational LLaVA models; §3 discusses computational challenges; §4 outlines our proposed solutions; §5 presents our empirical results and analysis; §6 offers further analysis and ablations; and §7 along with §8 survey related works and conclude the study, respectively.

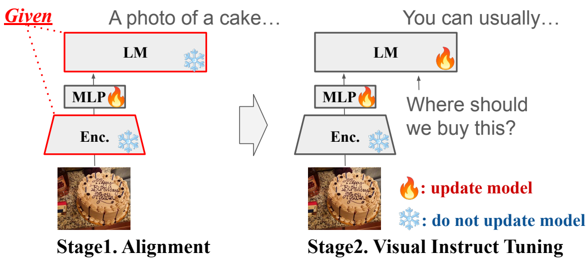


Figure 2: **Training pipeline** consists of two stages: alignment of visual and textual features through the MLP, followed by joint training of the LM and the MLP.

## 2 Large Language and Vision Assistants

**Architecture.** The LLaVA framework (Figure 2) employs a pre-trained Vision Transformer (ViT) as its vision encoder. Input images are resized and divided into patches of size $n \times (p_h \times p_w \times c)$, where $n = (h/p_h) \times (w/p_w)$. These patches are processed by the encoder to generate embeddings $\{\mathbf{z}_i \in \mathbb{R}^d\}$, which are then adjusted using a multi-layer perceptron (MLP) for compatibility with the language model. The AnyRes mechanism (Liu et al., 2024b) allows processing of larger images by segmenting them into $m$ parts. Each part is processed through the ViT, local embeddings are generated, and all are utilized. This ensures comprehensive understanding of varied image sizes and resolutions. However, this increases the total token count to $n \times m$, posing a computational challenge.

**Training Objectives and Datasets.** LLaVA is trained to minimize Cross-Entropy (CE) loss. During pre-training, it generates captions for images, with CE loss computed on the text. In the fine-tuning stage, given an image, question, and answer from the assistant, the loss is computed on the answer text. The LLaVA-1.5 dataset (Liu et al., 2023b) is widely used and this study aims to further enhance the dataset. More details are in Section 4.

## 3 Efficiency Challenges in LLaVA Models

This section addresses common overhead issues in LLaVA models, identifying critical limitations and defining the problem space for future work (Liu et al., 2024b; Dong et al., 2024).

### 3.1 Inference Overhead Sources

Inference overhead in LLaVA models stems from several factors:

- **Model Scale:** Larger models (e.g., 34B parameters) offer enhanced capabilities but incur significant computational costs.

| Model | Token Usage (#tok) | s/img | Memory (GB) |
|---|---|---|---|
| LLaVA-1.5-7B | 576 | 0.46 | 15 |
| LLaVA-1.5-13B | 576 | 0.66 | 28 |
| LLaVA-NeXT-7B | approx. 1.7–2.9K | 1.01 | 20 |
| LLaVA-NeXT-13B | approx. 1.7–2.9K | 1.78 | 40 |
| LLaVA-NeXT-34B | approx. 1.7–2.9K | 4.00 | 90 |

Table 1: Inference latency and memory costs for different LLaVA models. Tested with NVIDIA V100 GPUs.

- **Vision Encoder Complexity:** Advanced image encoders like SO400M and ViT-G (1.8B) improve performance but increase overhead (Sun et al., 2024; Zhai et al., 2023).
- **Image Resolution:** High resolutions (e.g., 900px) for detailed visual tasks like *DocVQA* (Mathew et al., 2021) increase computational demands on the vision encoder.
- **Vision Token Quantity:** Higher resolutions lead to more vision tokens, increasing the computational load on the LLM (e.g., LLaVA-NeXT uses up to 2880 tokens).

Higher image resolutions and complex tasks further increase computational demands on vision encoders and language models.

### 3.2 Benchmarking Baseline Models

Resource usage during inference for LLaVA and LLaVA-NeXT models was evaluated on the DocVQA and ChartQA (Masry et al., 2022) test sets. As shown in Table 1, LLaVA-1.5 models showed manageable computational costs, operable on a single V100 GPU. However, LLaVA-NeXT models with up to 2.9K tokens presented significant challenges. Testing on an NVIDIA V100 32GB showed that LLaVA-NeXT-13/34B could not be accommodated on a single GPU. These findings emphasize the challenges of larger models, especially in resource-constrained environments.

### 3.3 Existing Approaches to Efficiency

Existing computationally demanding methods come with trade-offs. Sampler modules like the Perceiver resampler (Alayrac et al., 2022) reduce token count but are resource-intensive to train and may struggle with generating precise responses (Liu et al., 2024b). Separate models for detailed or concise responses complicate deployment (Bai et al., 2024; Laurençon et al., 2024).

Simpler approaches, such as those based on LLaVA (Liu et al., 2023c), are compatible with libraries like *SGLang* (Zheng et al., 2023b) and *transformers* (Wolf et al., 2020), facilitating broader use. Improving these models for speed and performance remains a high-impact research area. For more on current trends, see Section 7.

## 4 Efficient Language and Vision Assistant

### 4.1 Preliminary: Base Architecture Modification and Initial Data Curation

To identify the most effective model architecture, we test various LLMs ranging from 160M to 13B as follows: Llama-160M, TinyVicuna-1B, Phi3-3.8B, Vicuna-7B, and Vicuna-13B.[2]

For the vision encoder, we replace the OpenAI CLIP-Large-336-14 module (LLaVA-1.5) with OpenAI CLIP-Base-224-32, using the Anyres technique to retain resolution while summarizing input into similar token count. OpenAI CLIP-Large-336-14 processes a 336x336 area into 576 tokens; OpenAI CLIP-Base-224-32 processes a 224x224 area into 49 tokens. Using Anyres, resolution increases to 896x676px with a token count capped at 637, compared to LLaVA-NeXT's 2880 tokens for 672x672px. However, this results in slight performance degradation (from LLaVA-1.5-7B's 40.1 to 39.8; see §4.1.1 in Figure 1).

We then expand the dataset to enhance performance by incorporating data from LLaVAR (Zhang et al., 2023), Idefics2 (Laurençon et al., 2024), and several open-source datasets. This involves using 1.1 million samples for alignment tasks and 1 million samples for instruction tuning. Detailed dataset quantities are provided in the Appendix. As shown in Figure 1, this strategy is somewhat effective (39.8 to 50.7) but does not achieve the performance levels of LLaVA-NeXT-7B which reaches 53.2 (refer to §4.1.2 in Figure 1).

### 4.2 Problem Definition and Strategies

Despite multiple optimizations, the model displayed performance issues, particularly generating hallucinations—incorrect responses due to inherent bias rather than accurate visual interpretation. These problems were evident in tasks requiring strong integration of visual and text information.

**Hypothesized Challenges.** We hypothesize two main challenges: (1) inadequate embeddings from

[2] Llama-160M: https://huggingface.co/Felladrin/Llama-160M-Chat-v1, TinyVicuna: https://huggingface.co/Jiayi-Pan/Tiny-Vicuna-1B, Phi3-3.8B (Abdin et al., 2024): https://huggingface.co/microsoft/Phi-3-mini-4k-instruct, Vicuna-7/13B (Zheng et al., 2023a): https://huggingface.co/lmsys

the vision encoder, and (2) poor understanding of basic text comprehension tasks, which is crucial for complex document interpretation.

**Improvement Strategies.** To address the challenges, we implemented: (1) a more efficient vision encoder to improve image embeddings quality, and (2) a training regimen prioritizing text comprehension before advancing to complex tasks.

To test our hypotheses, we conduct a series of comprehensive ablation experiments. Figure 1 illustrates our development stages. We tracked the effectiveness of our model modifications using various text-centric evaluation benchmarks, including DocVQA (**Doc**) (Mathew et al., 2021), ChartQA (**Chart**) (Masry et al., 2022), InfographicVQA (**Info**) (Mathew et al., 2022), and SEED-2-Plus (**SD2P**) (Li et al., 2024a). Additionally, we employed widely-used general LVLM benchmarks such as SEED (**SD-I**) (Li et al., 2023), MM-Star (**MMS**) (Chen et al., 2024), ScienceQA-IMG (**SciQA**) (Lu et al., 2022), and HallusionBench (**Hall**) (Guan et al., 2024). Our primary objective was to enhance performance on text-focused tasks while maintaining competitive performance on general tasks and ensuring low inference costs.

In the followings, we introduce each proposed module in detail and conduct extensive ablation studies, analyzing the impact of removing each component from the final model configuration.

### 4.3 Developing an Enhanced Vision Encoder with Weight Averaging

To improve visually-situated NLU, we have developed a new vision encoder optimized specifically for reading text within images. This encoder was designed to enhance resolution and computational efficiency while managing cost constraints.

Initially, we found that simply unfreezing the vision encoder while training the VLM (**C2**) did not yield significant improvements in performance. Next, we adopted a two-step approach. (1) We unfroze the vision encoder and trained it on a small VLM (1B scale) using text-centric datasets such as OCR-IDL (Biten et al., 2022). This training focused on a "text reading task," where the model was tasked with reading text within images. This allowed the vision encoder to adapt and enhance its text recognition capabilities. (2) We extracted the enhanced vision encoder (*REncoder*) from this text-centric VLM. Notably, the text-centric VLM used for achieving *REncoder* was not utilized further

| Vision Encoder Configuration | Text-Centric | General | Overall |
|---|---|---|---|
| **C1.** CLIP-B-224-AnyRes (CLIP) | 40.3 | 54.5 | 47.4 |
| **C2.** Unfreeze CLIP | 34.1 | 47.6 | 40.9 |
| **C3.** *REncoder* (RE) | 45.2 | 52.2 | 48.7 |
| **C4.** Avg (CLIP& RE) | 45.6 | 54.4 | 50.0 |
| **C5. ELVA-encoder** (Avg (CLIP& 12 REs)) | 45.7 | 54.7 | **50.2** |
| *Supplementary ablations* | | | |
| **C6.** CLIP-L-336 (*LLaVA-1.5 on our datasets*) | 37.5 | **58.6** | 48.1 |
| **C7.** CLIP (7%) + *REncoder* (93%) | **45.9** | 53.5 | 49.7 |

Table 2: Ablation Study Results for Different Vision Encoder Configurations. Average scores for text-centric tasks (DocVQA, ChartQA, InfoVQA, and SEED-2-Plus), and general image tasks (SEED, MMStar, ScienceQA, and HallusionBench) are reported. These results are obtained with Phi-3 (3.8B). The overall scores for other scales (from 1B to 13B) are shown in Figure 3.

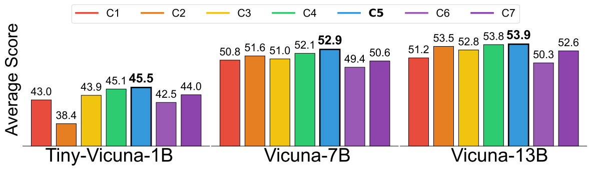


Figure 3: Performance of various vision encoder configurations at scales of 1B, 7B, and 13B. Average scores for each configuration (C1 to C7) across 8 benchmarks.

in our processes. When we trained a VLM with the obtained *REncoder*, we observe significant improvements in text-centric tasks (**C3**), however, its performance on general image tasks still declined.

Now we have two expert encoders, the original CLIP for general tasks and the *REncoder*. Inspired by prior work, *Model soup* (Wortsman et al., 2022), we tested with averaging the weights of the original encoder and the *REncoder*. Surprisingly, this method produced promising results (**C4**). Additionally, further adjusting the weight averaging ratios to lean more towards the *REncoder* provided marginally better text-centric performance (**C7**).

To further enhance robustness, we trained 12 *REncoder*s with different random seeds and then averaged their weights, a practice inspired by Wortsman et al. (2022). This averaging process, taking approximately 1.7 days on 8 V100 GPUs per phase, yielded a vision encoder that substantially improved text comprehension while maintaining robust general capabilities (**C5**). The additonal training details are in the Appendix.

In summary, the core idea is simple. We **(1) unfreeze the encoder and train a small VLM for text reading tasks, and retrieve the specialized encoder**, and **(2) make it robust to various tasks by applying weight averaging**. Finally, the produced vision encoder is used to build an efficient language and vision assistant, ELVA. Our new

**ELVA-encoder** brings substantial enhancements in text-centric tasks compared to the original base (**C1**). While there is still a reduction in general image performance compared to merely training LLaVA-1.5 on our data (**C6**), understanding the trade-offs in **C6** is key to fully appreciating the balance we achieve. We effectively balance overall performance and computational cost within the scope of CLIP-Base parameters (88M). The ELVA-encoder configurations demonstrate notable success overall, as shown in Table 2 and Figure 3.

### 4.4 Augmenting Text Understanding in Images with Read-and-Reason Prompts

We explore visual document understanding models utilizing pixel-based language modeling tasks, such as exhaustive image text reading during training (Kim et al., 2022; Lee et al., 2023). Models like LLaVA (Liu et al., 2023c,b) use OCR to preprocess images, augmenting user queries with OCR results. However, a comprehensive investigation of methods for supervising textual information during visual instruction tuning is lacking.

Table 3 shows ablation studies investigating text reading tasks during visual instruction tuning. Set **R1** follows standard practices using datasets without additional text reading components. In **R2**, inspired by prompting (Brown et al., 2020), we incorporate an initial QA task, *"What is written in this image?"* before QA on text-rich images. For example, with an image of a restaurant menu, the model first **reads all text before querying** about menu items or prices. This incremental addition improves performance significantly from **R1 to R2**, especially in text-rich tasks. We annotate datasets using OCR engines for this purpose.

Our further explorations assess this approach in resource-scarce environments, using 10% of the original instruction tuning dataset size. Figure 4 shows results across 1B to 13B parameter models. We also explore the supervision structure's impact by comparing "Read and Reason" versus "Reason and Read" approaches. **R3** models perform text reading last to evaluate this. Results confirm that "Read and Reason" is more effective, emphasizing structured prompting's importance in model learning. Lastly, we evaluate the effect of providing read text as context without explicit supervision (**R4**). Explicit supervision with text information yields marginal improvements in text-centric tasks.

In summary, the proposed core idea is to use RR-Prompt during model training to enhance text understanding in images. This approach, validated through ablation studies, shows significant performance improvements, especially in text-rich tasks. Note that the **RR-Prompt is used during training**; during inference, the model directly engages in reasoning, leveraging the enhanced capabilities acquired through the RR-Prompt, ensuring efficiency without needing an explicit text reading stage.

| | 1B Text/Gen/All | 3.8B Text/Gen/All | 7B Text/Gen/All | 13B Text/Gen/All |
|---|---|---|---|---|
| **R1.** | 41.6/48.7/45.2 | 43.7/53.9/48.8 | 47.5/56.0/51.8 | 48.5/56.5/52.5 |
| **R2.** | **42.4**/48.6/**45.5** | **45.7/54.7/50.2** | **49.2/56.6/52.9** | **50.4**/57.4/**53.9** |
| *Supplementary ablations* | | | | |
| **R3.** | 41.2/46.5/43.9 | 45.4/53.7/49.6 | 47.0/55.3/51.2 | 50.8/56.1/53.5 |
| **R4.** | 42.0/**48.8**/45.4 | 44.1/54.2/49.1 | 48.2/56.0/52.1 | 49.6/**57.6**/53.6 |

Table 3: Comparison of different model sizes and RR-Prompt variants. **R1** represents standard models trained without additional text reading prompt. **R2** employs explicit initial text reading steps for text-rich taks. **R3** carries out text reading at the end, while **R4** provides OCR results just as context without explicit supervision.

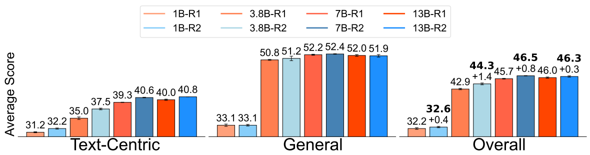


Figure 4: Using a 10% subset of the dataset, the impact of integrating the RR-Prompt during training is shown.

### 4.5 Bringing It All Together

To develop a more robust model capable of handling a wider range of tasks, we scaled up the model development by incorporating diverse datasets beyond merely text-centric tasks. Our final model integrates four additional datasets: Vision-Flan (Xu et al., 2024), RefCOCO (Kazemzadeh et al., 2014), VG (Krishna et al., 2017), and CORD (Park et al., 2019). By incorporating these additional datasets, we aimed to enhance both the performance and generalizability of our model. The final training involved 11K steps with a batch size of 128. The specific dataset details and schedules are in the appendix. As demonstrated in Figure 1, our final configuration shows solid performance.

## 5 Experimental Assessment

In this section, we rigorously test and evaluate our ELVA models under varying conditions. We aim to understand their capabilities and limitations by benchmarking them against baseline models across

| Model | # Param | | #tok | s/img | vram | *Text-Centric Benchmarks* | | | | | *General Multimodal Benchmarks* | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Vision | LM | | | | Doc | Chart | Info | SD2P | PBen | SD-I | MMS | SciQA | Hall | AI2D | Math | LBen |
| LLaVA-v1-13B | 300M | 13B | 576 | 1.43 | 26.9 | 9.8 | 7.0 | 19.9 | 39.5 | 14.0 | 51.2 | 33.0 | 62.4 | 43.0 | 43.9 | 25.9 | 69.9 |
| LLaVA-1.5-7B | 300M | 7B | 576 | 0.46 | 14.7 | 22.8 | 17.8 | 22.4 | 41.2 | 17.9 | 65.9 | 33.1 | 69.2 | 48.5 | 55.6 | 25.6 | 59.6 |
| LLaVA-1.5-13B | 300M | 13B | 576 | 0.66 | 27.0 | 24.5 | 18.5 | 24.9 | 44.4 | 19.6 | 68.2 | 34.1 | **72.3** | 45.7 | 60.7 | 27.7 | 66.1 |
| LLaVA-NeXT-7B | 300M | 7B | 1728-2880 | 1.01 | 20.0 | 68.3 | 51.9 | 31.6 | 51.7 | 49.6 | 69.8 | 38.2 | 69.0 | 44.8 | 66.8 | 31.8 | **72.3** |
| LLaVA-NeXT-13B | 300M | 13B | 1728-2880 | 1.78 | 40.1 | 69.8 | 59.0 | **34.9** | **55.6** | 57.3 | **71.5** | 41.2 | 73.4 | 46.7 | 71.7 | 34.1 | **72.3** |
| **ELVA-0.2B (ours)** | 88M | 0.2B | 98-637 | 0.24 | 1.4 | 44.7 | 50.3 | 14.8 | 31.4 | 12.3 | 37.8 | 31.5 | 39.0 | 48.1 | 31.0 | 27.0 | 28.4 |
| **ELVA-1B (ours)** | 88M | 1B | 98-637 | 0.41 | 3.3 | 62.6 | 57.7 | 23.7 | 36.8 | 27.3 | 52.3 | 32.6 | 63.3 | 50.4 | 46.9 | 31.7 | 36.0 |
| **ELVA-3.8B (ours)** | 88M | 3.8B | 98-637 | 0.45 | 9.1 | 66.5 | 62.5 | 24.3 | 43.4 | 24.4 | 61.1 | 35.9 | 73.4 | 53.7 | 64.2 | 34.9 | 40.7 |
| **ELVA-7B (ours)** | 88M | 7B | 98-637 | 0.54 | 14.5 | 69.1 | 61.8 | 30.7 | 47.7 | 45.0 | 62.6 | 35.4 | 74.7 | **56.8** | 66.2 | 36.6 | 50.7 |
| **ELVA-13B (ours)** | 88M | 13B | 98-637 | 0.72 | 27.0 | **71.7** | **65.2** | 34.6 | 52.6 | **59.2** | 65.3 | 41.2 | 77.7 | **56.8** | **69.3** | **38.1** | 51.0 |
| *Supplementary baselines* | | | | | | | | | | | | | | | | | |
| QwenVL-7B | 1882M | 7B | 224 | 0.50 | 19.2 | 65.1 | 60.2 | – | 41.0 | – | 56.5 | 34.0 | 61.6 | 37.4 | 57.2 | 15.5 | 12.9 |
| QwenVL-7B-Chat | 1882M | 7B | 224 | 0.56 | 19.2 | 62.6 | 49.3 | – | 46.9 | – | 62.9 | 34.0 | 64.0 | 40.8 | 59.7 | 34.9 | 67.7 |
| PaliGemma-3B | 428M | 3B | 1024 | 0.98 | 10.3 | – | 33.8 | – | 49.8 | – | 70.0 | **48.6** | **94.5** | 53.0 | **69.3** | 28.7 | 36.9 |

Table 4: **Performance Comparison Across Different Models and Benchmarks.** This table summarizes model sizes (Vision and LM), token counts (#tok), latency (s/img), and memory cost (vram). The performance metrics across various benchmarks are presented, showcasing each model's strengths and weaknesses in different challenges.

both text-based and image-based tasks.

### 5.1 Framework

Our evaluation process extends beyond our initial eight datasets, utilized in our ablation studies (See §4.2). To further enrich our examination, we have included additional diverse datasets such as **AI2D** (Kembhavi et al., 2016), MathVista-TestMini (Lu et al., 2024) (**Math**), LLaVA-Bench (Liu et al., 2023c) (**LBen**), along with Parsing-Bench (**PBen**) proposed in this work.

### 5.2 Generated Scenario-Based Benchmarks

In our research, we identify a significant gap in datasets representing real-world user scenarios for document assistants. To address this, we create the following datasets. These datasets will be open-source, and more details are in the appendix:

**CORD-Instruct.** Building on the CORD dataset, which consists of Indonesian receipts and their JSON annotations, CORD-Instruct provides instructional sets for models to generate outputs in JSON, XML, or Markdown formats. We utilize the OpenAI GPT-3.5 API to create these instructional sets, ensuring the exclusion of any erratic samples.

**Parsing-Bench.** Inspired by the LLaVA-Bench and LLM-as-a-Judge (Zheng et al., 2023a), we develop PARSING-BENCH to address the limitations of existing benchmarks like LLaVA-Bench, which include limited document-related samples and do not sufficiently reflect real user needs. Figure 6 presents example cases and model predictions. To test the model's ability to extract information from new documents, we create this dataset using 30 images from Brazilian Identity Documents(Álysson Soares et al., 2020) and SROIE(Huang et al., 2019), which are not used during training.

### 5.3 Results

Table 4 presents a detailed comparison of our ELVA models against baselines like PaliGemma (Beyer et al., 2022), QwenVL (Bai et al., 2024), and LLaVA across various benchmarks and tasks. The results, either reproduced or sourced from original papers, have been validated using VLMEvalKit (Contributors, 2023) and the official code by Liu et al. (2023c).

The ELVA models consistently demonstrate strong performance in text-centric and general image benchmarks. The ELVA-0.2B model, despite its smaller parameter count, performs admirably across various tasks. Larger models like ELVA-1B, ELVA-3.8B, and ELVA-7B exhibit superior performance, showcasing the benefits of increased model capacity. Notably, ELVA-7B strikes a fine balance between size and effectiveness. Our largest model, ELVA-13B, achieves top scores in benchmarks such as Doc, Chart, Hall, AI2D, Math, and PBen, underscoring the efficacy of scaling model capacity for vision-and-language tasks. Additionally, ELVA-13B remains efficient in terms of latency, reinforcing its practicality for diverse applications.

A notable limitation is observed in the LLaVA-Bench, where ELVA models underperform compared to the LLaVA series. Given this dataset only

| Method | s/img | vram | Doc | Chart | Info | SD2P |
|---|---|---|---|---|---|---|
| LLaVA-NeXT-7B | 1.01 | 20.0 | 68.3 | 51.9 | 31.6 | 51.7 |
| – w/ max. 1728 tokens | 0.70 | 17.1 | 51.7 | 48.0 | 27.9 | 44.9 |
| **ELVA-7B (ours)** | 0.54 | 14.5 | 69.1 | 61.8 | 30.7 | 47.7 |
| LLaVA-NeXT-13B | 1.78 | 40.1 | 69.8 | 59.0 | 34.9 | 55.6 |
| – w/ max. 1728 tokens | 1.11 | 30.4 | 53.9 | 52.3 | 30.9 | 49.2 |
| **ELVA-13B (ours)** | 0.72 | 27.0 | 71.7 | 65.2 | 34.6 | 52.6 |

Table 5: Ablations on reduced vision token counts.

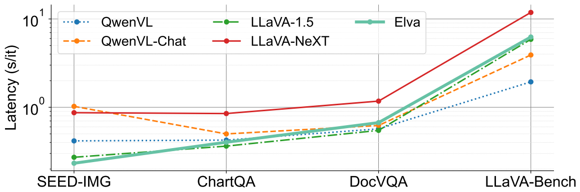


Figure 5: Latency comparison across multiple datasets.

comprises 24 images, interpretation requires caution. Detailed analysis will be provided in §6.3.

In summary, ELVA models deliver robust performance across a wide range of tasks and benchmarks. While increased model capacity generally enhances performance, efficiency and latency considerations are crucial for practical deployment. Our results underscore ELVA's proficiency in efficiently addressing diverse vision-and-language challenges, providing valuable insights into their performance and reliability across various contexts.

# 6 Further Analyses and Discussions

## 6.1 Ablations with LLaVA-NeXT Variants

To test the impact of reducing the number of tokens in LLaVA-NeXT models, we constrained the grid size, resulting in a maximum token count of 1728 (either 336x672 or 672x336 pixels). This adjustment speeds up processing while allowing us to evaluate performance changes across benchmarks. As shown in Table 5, reducing the vision token count leads to significant performance drops across all evaluated tasks. For example, the performance of the 13B model on DocVQA decreases from 69.8 to 53.9 when the token count is restricted, with similar trends observed in other variants.

In contrast, ELVA models demonstrate strong performance along with improved efficiency in both speed and memory usage, underlining their robustness in handling vision-and-language tasks efficiently. This analysis highlights the trade-off between token count and model performance: while reducing tokens can enhance computational efficiency, it may lead to a compromise in accuracy. The ELVA models effectively balance performance and efficiency, outperforming larger LLaVA-NeXT configurations even with reduced token counts.

## 6.2 Discussion on Memory and Time Costs

We evaluated latency in seconds per sample for benchmarks like ChartQA and DocVQA, as these tasks relate closely to real-world document information extraction scenarios. Zero-shot evaluations like SEED are less indicative of actual user scenarios, complicating meaningful latency comparisons. Task-oriented benchmarks like DocVQA offer more user-centric metrics. Benchmarks requiring longer answers, such as LLaVA-Bench, show inconsistent results due to varied answer lengths across models. Consequently, we focus on ChartQA and DocVQA for latency assessments but also include SEED and LLaVA-Bench results in Figure 5. These findings indicate ELVA maintains commendable latency across varied contexts.

While this study primarily focuses on inference time costs, training costs are equally important for practitioners. Despite processing a substantial dataset, our lightweight vision encoding facilitates high training sample throughput. Our training time costs are approximately 1.38 to 1.8 times that of LLaVA-NeXT, representing a reasonable trade-off given the efficiency and performance gains. Our approach remains competitive and more resource-efficient compared to several contemporary models. For detailed analysis, please refer to Appendix A.

Regarding memory usage, practical deployment often uses 4-bit quantization (Dettmers et al., 2023), significantly reducing memory costs. For example, the LLaVA-NeXT-13B model originally required two V100 GPUs but can run on a single V100 with quantization, though with increased latency and performance degradation. Despite limitations, quantization is promising and likely to improve with advancements. Our ELVA models, designed for efficiency, complement these advancements, promising even greater value when combined with quantization techniques. Future work will likely enhance these models' efficacy in deployment.

## 6.3 Case Study on LLM-as-a-Judge Sets

ELVA models perform robustly across benchmarks, excelling in text-centric tasks, but show performance dips on LLaVA-Bench. These issues often stem from failures in entity recognition, resulting in logically correct but penalized responses. For example, when asked to name an anime charac-

What is the name of the character?

The name of the character is not visible in the image. The image shows a character from an anime or manga, but the name of that character is unfortunately not visible. The image only shows the character's appearance, which includes a boy with brown hair, blue eyes, and a red bowtie (...)

**Elva received 5/10**

What is the name of the man in the photo?

I'm sorry, I cannot provide the name of the man in the photo. (...) the image does not provide any identifiable information about the man's name or any context that would allow me to provide it. The focus of the image is on the man and the teddy animal, and there is no visible text or context that could help identify the man.

**Elva received 2/10**

Can you summarize the purchased item information, including the name, price, and quantity, in XML format?


Certainly, here is the purchased item information in XML format:
```xml
&lt;purchased_items&gt;
&lt;item&gt;
&lt;name&gt;GREEN TEA&lt;/name&gt;
&lt;quantity&gt;1&lt;/quantity&gt;
&lt;price&gt;3.00&lt;/price&gt;
&lt;/item&gt;
&lt;item&gt;
&lt;name&gt;GREEN TEA&lt;/name&gt;
&lt;quantity&gt;1&lt;/quantity&gt;
&lt;price&gt;3.00&lt;/price&gt;
&lt;/item&gt;
&lt;item&gt;
&lt;name&gt;BUFFET DINNER (ADULT)&lt;/name&gt;
&lt;quantity&gt;3&lt;/quantity&gt;
&lt;price&gt;49.90&lt;/price&gt;
&lt;/item&gt;
&lt;/purchased_items&gt;
```

**Elva received 9/10**

Figure 6: Results on LLaVA-Bench and Parsing-Bench.

ter, ELVA scored 5/10 for failing to provide the correct name despite a detailed description. Similarly, it scored only 2/10 in identifying a person in a photo without textual clues. These dips highlight limitations in the ELVA-encoder's (88M) entity memorization capacity and the lack of contextual data within images. Adding targeted data during instruct-tuning might address these issues, though feasibility is debatable. On the other hands, ELVA excels in information comprehension tasks, such as summarizing receipt data into XML format, scoring 9/10. This shows its ability to accurately read, comprehend, and organize text information. While ELVA is efficient, it remains competitive to other models rather than significantly superior. Future research may focus on enhancing the vision encoder and improving contextual understanding to overcome these limitations.

### 6.4 Discussion on Leveraging OCR

Incorporating OCR can be effective for handling text-rich high-resolution images. For instance, Kim et al. (2023) utilized OCR in their VLM for Visual Document Understanding (VDU) tasks, suggesting similar applications for models like LLaVA or ELVA. We integrated OCR outputs as context during inference, as shown in Table 6, resulting in

| Method | Doc | Chart | Info | SD2P |
|---|---|---|---|---|
| LLaVA-NeXT-7B | 74.5 (↑6.2) | 53.7 (↑1.8) | 35.5 (↑3.9) | 55.3 (↑3.6) |
| **ELVA-7B (ours)** | **77.8** (↑8.7) | **64.0** (↑2.2) | **39.5** (↑8.8) | **55.7** (↑8.0) |
| LLaVA-NeXT-13B | 76.5 (↑6.7) | 62.5 (↑3.5) | 40.4 (↑5.5) | 58.9 (↑3.3) |
| **ELVA-13B (ours)** | **81.1** (↑9.4) | **67.5** (↑2.3) | **44.8** (↑10.2) | **60.6** (↑8.0) |

Table 6: Performance improvement with OCR.

marked improvements, especially for ELVA. However, OCR processing has costs. Using the CLOVA OCR API[3], our tests on the DocVQA dataset averaged about 4 seconds per sample. Faster OCR engines exist but often at the expense of accuracy. Additionally, upscaling VLMs to handle very high resolutions (e.g., 4K, 8K) may not be practical. Thus, leveraging OCR and similar tools remains a valuable area of exploration, aiming to balance specialized tools and VLMs for optimal performance.

## 7 Related Work

**Visually-Situated NLU** has evolved significantly, starting with heavy reliance on OCR for text extraction. A major advancement was marked by Xu et al. (2020), integrating textual and layout information for sophisticated document understanding. The field progressed to OCR-free models (Kim et al., 2022; Lee et al., 2023), enhancing flexibility and accuracy. Further, **LVLMs** aim to jointly process visual and textual data, enhancing cross-modal understanding. LLaVA (Liu et al., 2023c) demonstrate efficient cross-modal performance. In VDU, Kim et al. (2023) use a hybrid approach, balancing OCR-free and OCR-dependent modes. More recently, high-resolution processing is central to models (Liu et al., 2024b; Dong et al., 2024), which achieve superior results despite computational demands. For additional comparisons with recent models, refer to Appendix A.

## 8 Conclusion

This study introduces ELVA, a robust and efficient model framework excelling in text-centric and visual tasks. Our empirical studies show ELVA models consistently outperform existing baselines, achieving notable efficiency in memory usage and latency. Challenges in entity recognition and contextual understanding highlight areas for improvement. Future research should focus on refining these aspects to enhance ELVA's performance and robustness, ultimately solidifying its position as a leading solution in visually-situated NLU.

[3] https://clova.ai/ocr/en

## 9 Limitations

Despite the significant advancements demonstrated by ELVA, several limitations remain. Firstly, ELVA occasionally struggles with recognizing specific entities and fully understanding contextual nuances within images, leading to penalized responses despite logical accuracy. That is, while ELVA excels in text-centric tasks, it may have limitations in the vision encoder's entity memorization and contextual data handling.

Incorporating OCR is beneficial for processing text-rich, high-resolution images, but it introduces latency and potential accuracy trade-offs. Moreover, managing very high-resolution images (4K or 8K) still remains challenging due to the balance required between performance improvements and computational resources.

While ELVA achieves lower inference costs and maintains a reasonable training time, the substantial data volume processed can lead to moderate time differences, as discussed in Appendix A. This underscores the importance of continued optimization in both training efficiency and performance.

Future research should focus on refining entity recognition, contextual understanding, training efficiency, and OCR handling. Investigating the balance between specialized tools like OCR and the core VLM is essential for optimizing performance. Additionally, expanding ELVA's capabilities to handle multilingual tasks will further increase its applicability and utility.

## 10 Ethical Considerations

Developing ELVA involves important ethical responsibilities such as reducing data biases and ensuring transparency. To manage these, we use only controlled and verified open-source datasets for model training. Currently, we rely on the autoregressive models' direct output, but we could also use post-processing techniques or additional training methods to address biases and privacy issues better. By open-sourcing our models and datasets, we encourage peer reviews and collaboration to solve ethical challenges, promoting accountability. These steps help ensure that ELVA upholds high ethical standards and is used for beneficial purposes while minimizing risks.

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# A Consideration on Model Training Cost and Additional Comparison to Recent Models

Table 7 summarizes the training costs for different model sizes of ELVA using 8 A100 GPUs. Regarding LLaVA-NeXT's training costs, estimates obtained from the official blog report[4] suggest that our training times are approximately 1.38 to 1.8 times longer (they reported needing 20 hours for a 7B model using 8 A100 GPUs and 24 hours for a 13B model using 16 A100 GPUs). While training time can vary depending on the measurement environment, the data indicates that our model's training costs are comparable despite its relatively small and lightweight nature. This is especially evident when considering the significantly larger data requirements of contemporary models like QwenVL (Bai et al., 2024), which uses 1.4B data points for pre-training and 50M for instruction tuning.

Furthermore, our observation in Section 4.3 shows that even with additional data, LLaVA-1.5 did not surpass the overall score we achieved with ELVA (**C5** vs. **C6**), strengthening the argument for ELVA's efficiency.

Although there could be concerns regarding the increased use of data and resources, placing our approach within the context of recent advancements offers a different perspective. For example, QwenVL's substantial resource investment in data and training time underscores the relatively modest yet effective approach of ELVA. Similarly, other models, such as Shikra (Chen et al., 2023a) and Idefics2 (Laurençon et al., 2024), demonstrate the different scales of resource utilization. Shikra uses 600K data points for alignment and 5.5M for instruction tuning, while Idefics2 achieves impressive results with over 1B data points.

In contrast, Monkey (Li et al., 2024b), while contributing significantly to the field, leverages the LLM from QwenVL for further training. This makes direct comparisons of the total training cost less relevant. Our primary focus is on scenarios where practitioners utilize readily available open-source LLMs and vision encoders to build their systems. Therefore, recent VLM works that adopt such an approach are excluded from direct comparisons.

Moreover, it is crucial to highlight the importance of inference cost. Methods focusing

[4] https://llava-vl.github.io/blog/2024-01-30-llava-next

| Model Size | Alignment Time | Instruct Tuning Time | Total Time |
|---|---|---|---|
| 0.2B | 0.5 hours | 3 hours | 3.5 hours |
| 1B | 1.5 hours | 6 hours | 7.5 hours |
| 3.8B | 4.5 hours | 18.5 hours | 23 hours |
| 7B | 6.5 hours | 29.5 hours | 36 hours |
| 13B | 11 hours | 55 hours | 66 hours |

Table 7: Training times for various model sizes on 8 A100 GPUs.

on reduced inference costs often face prohibitive training expenses and challenges in maintaining instruction-following capabilities across varied response lengths (Dai et al., 2023; Liu et al., 2023b; Laurençon et al., 2024). Among LLaVA-like simple architectures, ELVA stands out as a fast, lightweight, and cost-efficient alternative.

Furthermore, while there are ongoing efforts to develop small yet robust VLMs in the range of 1.7B to 3B parameters (Chu et al., 2024), these works predominantly focus on general image processing. It is important to note that there is still a lack of efficient VLMs specifically designed for visually-situated NLU tasks, such as processing document images in a high-resolution context, which is the primary focus of our work.

## B Implementation Details

### B.1 Software and Hardware Setup

Our experiments are performed using the official code of LLaVA (Liu et al., 2023c)[5]. We utilize NVIDIA V100 and A100 GPUs for the computations. Ablation studies are conducted on V100 GPUs, whereas the final configuration models run on A100 GPUs. We do not observe any significant performance difference based on the type of GPU used. However, training on V100 GPUs is approximately 2 to 3 times slower per step compared to A100 GPUs. Although our codebase is based on LLaVA, to ensure better reproducibility, we will release the scripts used for training our models and any necessary code modifications as open-source.

### B.2 Datasets and Hyperparameters

**Initial curated dataset: LLaVA-1.5** Table 8 provides detailed quantities of the dataset subsets.

**Datasets** Table 9 lists the datasets in Elva's final configuration. Meanwhile, for the alignment phase, we used the respective alignment datasets from LLaVA (Liu et al., 2023c) and LLaVAR (Zhang et al., 2023), which consist of 558K and 422K samples respectively.

[5] https://github.com/haotian-liu/LLaVA

| Dataset | # Samples |
|---|---|
| LLaVA | 157,712 |
| SG40k | 40,688 |
| VQA-v2 | 82,783 |
| GQA | 72,140 |
| OKVQA | 8,998 |
| OCRVQA | 80,000 |
| A-OKVQA | 66,160 |
| TextCaps | 21,953 |
| RefCOCO | 48,447 |
| VG | 86,417 |

Table 8: Dataset proportions for LLaVA-1.5 (Liu et al., 2023b). RefCOCO and VG were not used in ELVA ablation studies.

**Hyperparameters** Table 10 and Table 11 outline the hyperparameters applied during the alignment and instruction tuning stages, noting that smaller models thrive with larger learning rates. In the final model training configuration, we use the data and sampling ratios from Table 9 and train the model for 11K steps. The exact number of unique images is difficult to calculate since some images are shared across datasets, but we estimate using approximately 1M unique images. Note that LLaVA-NeXT reported using 760K samples,[6] indicating a modest increase in our case. Moreover, as we leverage multiple curated datasets with slightly different questions on the same images, we consider a synthetic epoch to consist of 1.4M examples. Thus, with a batch size of 128, we run 11K steps (1.4M / 128). For the Ablation studies, we exclude datasets like VG, RefCOCO, and Vision-Flan to reduce the cost, resulting in 9K training steps. Additionally, the 0.2B model converges more slowly, so we run this model for twice the number of steps compared to the other models (1B to 13B). This increased number of steps is applied only to the 0.2B model.

## C Details on the Construction of the Newly Generated Datasets

### C.1 CORD-Instruct

**Process of Creating the CORD-Instruct Dataset** CORD (Comprehensive OCR Dataset) provides Indonesian receipt data along with structured information represented in JSON format. Our goal was to generate a dataset that requires models to

[6] https://llava-vl.github.io/blog/2024-01-30-llava-next

| Dataset | # Samples | Sampling Ratio % |
|---|---|---|
| LLaVA-1.5-Set (Liu et al., 2023b) (See Table 8) | 665,298 | 46.49 |
| Vision-Flan-Set (Xu et al., 2024) | 186,103 | 12.99 |
| WikiArt (Chen et al., 2023b) | 500 | 0.03 |
| Celebrity (Chen et al., 2023b) | 498 | 0.03 |
| Landmark (Chen et al., 2023b) | 500 | 0.03 |
| Share-TextVQA (Chen et al., 2023b) | 500 | 0.03 |
| DocVQA (Mathew et al., 2021) | 11,480 | 1.60 |
| ChartQA (Masry et al., 2022) | 18,317 | 2.56 |
| *Cauldron-Set-AI2D | 2,434 | 0.17 |
| *Cauldron-Set-Chart2Text | 26,961 | 1.88 |
| *Cauldron-Set-Diagram-Image-to-Text | 300 | 0.02 |
| *Cauldron-Set-HITAB | 2,500 | 0.17 |
| *Cauldron-Set-IAM | 5,663 | 0.40 |
| *Cauldron-Set-RenderedText | 10,000 | 0.70 |
| *Cauldron-Set-Robut-SQA | 8,514 | 0.59 |
| *Cauldron-Set-Robut-WTQ | 38,246 | 2.67 |
| *Cauldron-Set-ScienceQA | 4,976 | 0.35 |
| *Cauldron-Set-Screen2words | 15,730 | 1.10 |
| *Cauldron-Set-STVQA | 17,247 | 1.20 |
| *Cauldron-Set-TabMWP | 22,722 | 1.59 |
| *Cauldron-Set-InfoVQA | 2,118 | 0.15 |
| *Cauldron-Set-TQA | 1,493 | 0.10 |
| CORD-Instruct (Proposed in this work, §5.2) | 680 | 0.05 |
| VisualMRC (Tanaka et al., 2021) | 7,959 | 0.56 |
| LLaVAR-Inst (Zhang et al., 2023) | 19,732 | 1.38 |
| DocReason (Hu et al., 2024) | 25,877 | 1.81 |
| DocVQA-single† | 44,815 | 9.38 |
| ChartQA-single† | 28,068 | 5.88 |
| Layout-en-sampled‡ (Kim et al., 2023) | 50,000 | 3.49 |
| DVQA-sampled‡ (Kafle et al., 2018) | 10,000 | 0.70 |
| MMC-Chart-sampled‡ (Liu et al., 2024a) | 10,000 | 0.70 |
| ScreenQA-sampled‡ (Hsiao et al., 2024) | 10,000 | 0.70 |
| LRV-Chart-sampled‡ (Liu et al., 2023a) | 6,746 | 0.47 |

Table 9: Overview of datasets used in ELVA's final data configuration. All datasets are open-source and freely accessible. Datasets marked with * are subsets curated by Laurençon et al. (2024), with only selected portions adopted in this work. † indicates that each question-answer pair is considered as a single sample. Datasets marked with ‡ had a large volume of data, hence, only partial images were randomly sampled.

| Model Size | LR | Epsilon | Grad Clip Norm | Weight Decay | Warmup Ratio |
|---|---|---|---|---|---|
| 0.2B | 1e-3 | 1e-6 | 0.5 | 0.0 | 0.03 |
| 1B | 1e-3 | 1e-6 | 0.5 | 0.0 | 0.03 |
| 3.8B | 1e-3 | 1e-6 | 0.5 | 0.0 | 0.03 |
| 7B | 1e-4 | 1e-6 | 0.5 | 0.0 | 0.03 |
| 13B | 1e-4 | 1e-6 | 0.5 | 0.0 | 0.03 |

Table 10: Hyperparameters used during the alignment stage.

produce outputs in specific formats such as JSON, XML, or Markdown.

The CORD-Instruct dataset was generated using the OpenAI GPT-3.5 API. We crafted a detailed prompt (as outlined in Table 13) to guide the API in generating synthetic user queries and corresponding responses based on the given structured information.

After generating initial instructions and OpenAI's response with synthetic questions and answers, we manually filtered and annotated the samples to ensure they accurately represented real-world scenarios. This included removing ambiguous instructions and correcting poorly transcribed JSON, XML, or Markdown outputs.

| Model Size | LR | Epsilon | Grad Clip Norm | Weight Decay | Warmup Ratio |
|---|---|---|---|---|---|
| 0.2B | 3e-4 | 1e-6 | 0.5 | 1e-3 | 0.03 |
| 1B | 3e-4 | 1e-6 | 0.5 | 1e-3 | 0.03 |
| 3.8B | 2e-4 | 1e-6 | 1.0 | 0.0 | 0.03 |
| 7B | 2e-5 | 1e-6 | 1.0 | 0.0 | 0.03 |
| 13B | 2e-5 | 1e-6 | 1.0 | 0.0 | 0.03 |

Table 11: Hyperparameters used during the instruct tuning stage. Larger learning rates were noted to be more effective for smaller models.

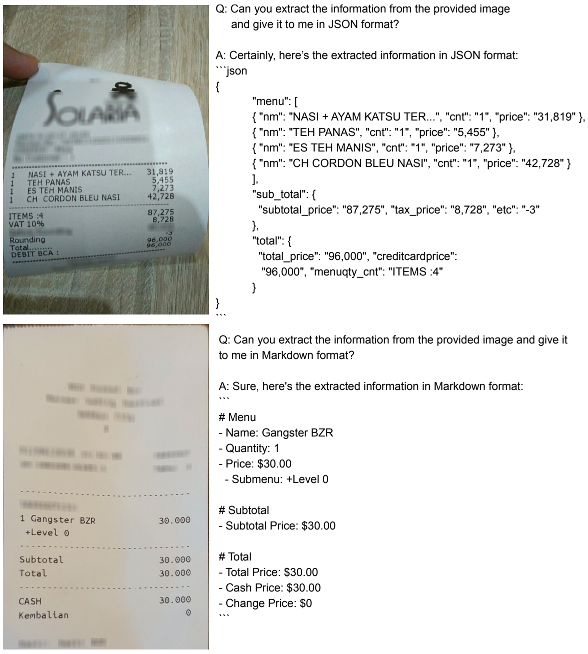


Figure 7: The generated examples of CORD-Instruct.

## C.2 Parsing-Bench

Parsing-Bench is developed to fill the gap in datasets that cater to the real-world needs of document assistants. Drawing inspiration from LLaVA-Bench, this benchmark encompasses tasks that demand intricate document interpretation and analysis.

Brazilian Identity Documents (BID) (Álysson Soares et al., 2020) and SROIE (Huang et al., 2019) are utilized in the creation of Parsing-Bench.

Tables 15 and Tables 16 shows the prompts we employ to extract contextual information from each image. To illustrate our approach, Figure 8 provides an example where the evaluation process is depicted in detail.

**Example Cases and Specific Details of the Dataset** Parsing-Bench includes examples requiring the model to extract and interpret information from various types of documents, such as tables, forms, and invoices. The data were carefully curated to challenge the models' comprehension and reasoning capabilities. Example cases and detailed

| Batch Size | Learning Rate (LR) | Weight Decay |
|---|---|---|
| 128 | 5e-5 | 1e-3 |
| 128 | 6e-5 | 1e-3 |
| 128 | 7e-5 | 1e-3 |
| 128 | 8e-5 | 1e-3 |
| 256 | 5e-5 | 1e-3 |
| 256 | 6e-5 | 1e-3 |
| 256 | 7e-5 | 1e-3 |
| 256 | 8e-5 | 1e-3 |
| 512 | 5e-5 | 0.0 |
| 512 | 6e-5 | 0.0 |
| 512 | 7e-5 | 0.0 |
| 512 | 8e-5 | 0.0 |

Table 12: The configurations for 12 difference *REncoder* training, including batch size, learning rate, and weight decay. Other hyperparameters basically identical to §B.2.

**CORD-Instruct Generation Prompt**

Create a synthetic user query requesting information extraction from a given document image. The extracted information should be provided in various formats such as JSON, XML, or Markdown based on the user's request. Users may ask for specific parts of the information, like menu items or payment amounts. Your task is to generate both a user query and the corresponding response from the information extraction system. Ensure the queries and responses vary in detail and format. Sometimes include concise responses, particularly when indicated with the word "concisely."
The provided JSON is to guide your response creation - do not display or mention it in the user queries. Specifically, do not create queries that ask for information extraction from a provided JSON (e.g., "Can you extract the information from the provided JSON" is not allowed). Additionally, you do not need to strictly follow the tag names in the provided JSON while creating your responses (e.g., "nm" can be "name" or "cnt" can be "count" ). Return results strictly in the format shown below:

```
Query: I need the payment amount
from the document in this image
in JSON format, answer concisely.
Answer: <<<json
{
  "payment_amount": "$123.45"
}
>>>

Query: Please parse the input document
and provide the menu details in XML.
Answer: Certainly, here is the menu
information in XML format:...

Query: Can you provide the extracted
customer information from the document
image in Markdown?
Answer: Sure, here it is:
<<<
# Customer Information
- Name: John Doe
- Email: john.doe@example.com
>>>
```

Remember, the goal is to include appropriate formatting such as JSON, XML, or Markdown in your responses to correspond with the user's query.

Table 13: CORD-Instruct Prompt for Data Generation

descriptions will be made publicly available.

## D ELVA-encoder Training

**Training Details** Each *REncoder* is trained by unfreezing the vision encoder and fine-tuning it on a small VLM (1B scale) with text-centric datasets. The training focuses on text reading tasks to enhance the vision encoder's text recognition capabilities.
We use the following datasets for training: OCR-IDL (Biten et al., 2022) (837,922 samples used), PDFA[7] (1,048,569 samples used), LLaVA (Liu et al., 2023c) alignment set, and LLaVAR (Zhang et al., 2023) alignment set. The model is trained with duplicate sampling allowed, treating 3.5M samples as a virtual epoch and conducting 1 epoch of training using the configuration below 12 times. Since the model has 1 billion parameters, the training proceeds relatively quickly, with each session consuming approximately 1.7 days using 8 V100 GPUs.

**Weight Averaging with Model Soup** Although Wortsman et al. (2022) proposed a greedy soup method involving validation where some models are averaged or rejected, we employed a uniform averaging technique referred to as "uniform soup" in the paper.

## E Prompts used in RR Prompts

As seen in Table 14, we insert a preliminary QA turn where the model is prompted to read the text in the image using simple commands. Although the idea is straightforward, we observe that it leads to better training outcomes for the model. To avoid specialization to a single OCR engine, we use a mix of MS OCR[8] and CLOVA OCR[9] annotations. In text-heavy samples like those in DocVQA, the CLOVA OCR API takes approximately 4 seconds per call. However, we believe this cost can be optimized in future iterations.

[7] https://huggingface.co/datasets/pixparse/pdfa-eng-wds

[8] https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/overview-ocr.

[9] https://clova.ai/ocr/en

| Prompt |
| --- |
| Carefully decipher the text in this image. Provide the text in the image only. |
| Investigate the image for any text. Provide the text in the image only. |
| Examine the image for any letters or words. Provide the text in the image only. |
| Identify all written characters present in the image. Provide the text in the image only. |
| Do a careful reading of the image and transcribe all text. Provide the text in the image only. |
| Inspect the image and write down all readable characters. Provide the text in the image only. |
| Translate the image content into written text. Provide the text in the image only. |
| Review the image and offer a transcription of the text. Provide the text in the image only. |
| Look over the image and jot down all visible text. Provide the text in the image only. |
| Scrutinize the image for any discernible words or letters. Provide the text in the image only. |
| Study the image and document all characters found within. Provide the text in the image only. |
| Assess the image and record any visible words or letters. Provide the text in the image only. |
| Kindly extract any text or characters from this image. Provide the text in the image only. |
| Evaluate the image and isolate all written content. Provide the text in the image only. |
| Assimilate all readable characters within the image. Provide the text in the image only. |
| Decode any legible text visible in the image. Provide the text in the image only. |
| Grasp all written elements within the image. Provide the text in the image only. |
| Conduct a thorough examination of the image and capture all text. Provide the text in the image only. |
| Peel all readable characters out from the image. Provide the text in the image only. |
| Interpret all characters from the image. Provide the text in the image only. |
| Understand and transcribe any textual content from the image. Provide the text in the image only. |
| Dissect the image and recover any written characters. Provide the text in the image only. |
| Deduce and present all visible text from the image. Provide the text in the image only. |
| Inspect the image and translate any visible characters into text. Provide the text in the image only. |

Table 14: **Prompt examples for text reading tasks used in Read-and-Reason Prompting**

**BID Context Generation Prompt**

Using the provided Brazilian Identity Document image, please compose a comprehensive and detailed caption that encapsulates all the elements depicted in the image. Ensure precision in extracting any text present, maintaining case sensitivity and retaining the exact original form. Begin with a well-written caption in natural language, detailing the image's content, layout, and nuances. Conclude with a well-structured XML format that meticulously documents the extracted information, preserving the image's original layout and details.

Table 15: Parsing-Bench Context Generation Prompt for BID Dataset

**SROIE Context Generation Prompt**

Using the provided scanned receipt image, please compose a comprehensive and detailed caption that encapsulates all the elements depicted in the image. Ensure precision in extracting any text present, maintaining case sensitivity and retaining the exact original form. Begin with a well-written caption in natural language, detailing the image's content, layout, and nuances. Conclude with a well-structured XML format that meticulously documents the extracted information, preserving the image's original layout and details.

Table 16: Parsing-Bench Context Generation Prompt for SROIE Dataset

Can you summarize the purchased item information, including the name, price, and quantity, in XML format?

ROCKU YAKINIKU
CPI ROCKU SDN. BHD. (1096317-P)
GST NO.: 000446271488
LOT F355/356/357,FIRST FLOOR,
RAINFOREST,1 UTAMA SHOPPING CENTRE,
LEBUH BANDAR UTAMA,
47800 PETALING JAYA

19/02/2018 17:43 T001 A063975
Csh: CASHIER Svr: BISWASH(ip
Table: A9 Pax: 3
BILL: 063975 RM

GREEN TEA
1 3.00 3.00 T
GREEN TEA
1 3.00 3.00 T
BUFFET DINNER (ADULT)
3 49.90 149.70 T
SUBTTL 155.70
10% SVC CHG 15.57
6% GST 10.28

TOTAL 181.55
CASH 201.55
CHANGE 20.00

| GST Summary | Amount | Tax |
| --- | --- | --- |
| T:6% GST | 171.27 | 10.28 |
| Z:Zero Rated | 0.00 | 0.00 |

Signature:________________
Thank you. Please come again
Bill Close: 19:08:04

Certainly, here is the purchased item information in XML format:

```xml
<purchased_items>
  <item>
    <name>GREEN TEA</name>
    <quantity>1</quantity>
    <price>3.00</price>
  </item>
  <item>
    <name>GREEN TEA</name>
    <quantity>1</quantity>
    <price>3.00</price>
  </item>
  <item>
    <name>BUFFET DINNER (ADULT)</name>
    <quantity>3</quantity>
    <price>49.90</price>
  </item>
</purchased_items>
```

**Elva received 9/10**

**REVIEW:**
**8 9**

Assistant 1 provided a clear and accurate summary of the purchased items in XML format. However, the XML tags used ("Receipt", "Item", "Name", "Price", "Quantity") are not consistent with the tags used in the original XML provided by the user. This inconsistency might cause confusion or require additional work to integrate with the user's existing data structure.

Assistant 2 also provided a clear and accurate summary of the purchased items in XML format. The tags used ("purchased_items", "item", "name", "quantity", "price") are more consistent with the original XML structure provided by the user, making it easier to integrate and more relevant. Additionally, Assistant 2's response includes a root tag ("purchased_items") that encapsulates all items, which is a good practice for XML structure.

Overall, both assistants performed well, but Assistant 2's response is slightly more aligned with the user's existing data structure and includes better XML formatting practices.

The image is a scanned receipt from ROCKU YAKINIKU, a restaurant located at CPI ROCKU SDN. BHD. (1096317-P), LOT F355/356/357, FIRST FLOOR, RAINFOREST, 1 UTAMA SHOPPING CENTRE, LEBUH BANDAR UTAMA, 47800 PETALING JAYA. The receipt includes the GST number 000446271488. The transaction took place on 19/02/2018 at 17:43, with the cashier identified as CASHIER and the server as BISWASH(ip). The table number is A9, and there were 3 people (Pax: 3). The bill number is 063975.

The items ordered include:
- 1 Green Tea at RM 3.00
- 1 Green Tea at RM 3.00
- 3 Buffet Dinner (Adult) at RM 49.90 each, totaling RM 149.70

The subtotal is RM 155.70. Additional charges include a 10% service charge of RM 15.57 and a 6% GST of RM 10.28, bringing the total to RM 181.55. The payment was made in cash, with RM 201.55 given, resulting in a change of RM 20.00.

The GST Summary shows:
- T: 6% GST on an amount of RM 171.27, with a tax of RM 10.28
- Z: Zero Rated with an amount of RM 0.00 and a tax of RM 0.00

The receipt concludes with a space for a signature and a thank you note, "Thank you. Please come again." The bill was closed at 19:08:04.

```xml
<receipt>
  <restaurant>
    <name>ROCKU YAKINIKU</name><company>CPI ROCKU SDN. BHD. (1096317-P)</company><gst_no>000446271488</gst_no>
    <address><lot>LOT F355/356/357, FIRST FLOOR</lot><location>RAINFOREST, 1 UTAMA SHOPPING CENTRE</location><street>LEBUH BANDAR UTAMA</street><city>47800 PETALING JAYA</city></address>
  </restaurant>
  <transaction>
    <date>19/02/2018</date><time>17:43</time><terminal>T001</terminal><bill_no>A063975</bill_no>
    <cashier>CASHIER</cashier><server>BISWASH(ip)</server><table>A9</table><pax>3</pax><bill>063975</bill>
  </transaction>
  <items>
    <item><name>GREEN TEA</name><quantity>1</quantity><price>3.00</price><total>3.00</total><tax>T</tax></item>
    <item><name>GREEN TEA</name><quantity>1</quantity><price>3.00</price><total>3.00</total><tax>T</tax></item>
    <item><name>BUFFET DINNER (ADULT)</name><quantity>3</quantity><price>49.90</price><total>149.70</total><tax>T</tax></item>
  </items>
  <summary><subtotal>155.70</subtotal><service_charge>15.57</service_charge><gst>10.28</gst><total>181.55</total>
    <payment><method>CASH</method><amount>201.55</amount><change>20.00</change></payment>
  </summary>
  <gst_summary>
    <gst><type>T</type><rate>6%</rate><amount>171.27</amount><tax>10.28</tax></gst>
    <gst><type>Z</type><rate>Zero Rated</rate><amount>0.00</amount><tax>0.00</tax></gst>
  </gst_summary>
  <footer>
    <signature>________________________</signature><note>Thank you. Please come again</note><bill_close>19:08:04</bill_close>
  </footer>
</receipt>
```

Figure 8: The top left displays the question and the model prediction results. The top right contains the received review. The bottom part, as introduced in Section C.2, exhibits the image's context prepared through prompting. The evaluation compares two model predictions by inputting them into a high-performance LLM along with this context. In this example review, ELVA is assistant 2.